Worst 2020s NBA Contract Analysis

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Introduction

Some students plan for their senior year of college to be arguably their lightest. Meanwhile, this fall semester of mine at the University of Illinois has already kicked into high gear despite beginning barely a month ago. Since then, I've been faced with tackling a plethora of job applications, student organization meetings, and, of course, classes. One could imagine how this easily leads to the need for an occasional break. So, I took one: I opened Instagram on my phone and after "doomscrolling" for only a short while, I came across a particular post.

Credit to Lucas Burns (@itslucasburns) on Instagram—on August 24th of this year, he posted an ordered compilation of what he believed to be the worst free agent contracts of the 2020s in the National Basketball Association. He landed on sixteen, explaining that the first half were "death sentences" while the latter half were "simply good players who were just unfortunately overpaid." This, of course, piqued my interest, which led me to question: to what extent was this list compiled using empirical statistical practices?

As a soon-to-be university graduate with a Bachelor's degree in Economics, I have my eyes set primarily on data analysis and consulting roles—the dream being basketball strategy—roles that many of my peers and I wonder about the extent to which they will be replaced by artificial intelligence. Irrespective of that concern, I saw this as an opportunity to display some of my skills, brush up on others, and, most importantly, have some fun. Just one question came to mind: can we purposefully measure ROI on NBA contracts down to the dollar?

Methodology

To answer whether we can purposefully measure ROI on NBA contracts, I first define an empirical valuation metric. Grounded in the sports adage that "the best ability is availability," the first independent variable I consider is Games Played (GP). Furthermore, I incorporate Annual Salary (Salary) and a third metric known as Player Impact Estimate (PIE). As explained by whatslaps in his breakdown of advanced basketball metrics, PIE measures a player's production against overall game context ("NBA Advanced Stats for DUMMIES"). The metric gauges the percentage of game events that a player contributed to in a single game or as an aggregate across an entire season. Historically, the player with the highest PIE rating at the end of a season frequently wins the league's Most Valuable Player award. PIE encompasses all standard box score metrics (points, rebounds, assists, steals, blocks, turnovers, and personal fouls) to capture two-way impact more holistically than traditional metrics like Player Efficiency Rating (PER).

Initially, I evaluated a player's annual output per dollar spent against their contract using durability and rate efficiency:

$$Player\ ROI\ =\ \frac{GP\ \cdot\ PIE}{Annual\ Salary\ (Millions)}$$

However, a critical adjustment was necessary: because PIE measures efficiency on a per-possession basis, a reserve logging 12 minutes per game could theoretically post the same PIE as a superstar playing 36 minutes. Multiplying by Minutes Per Game (MPG) accounts for true total workload:

$$Total\ Impact\ ROI\ =\ \frac{GP\ \cdot\ MPG\ \cdot\ PIE}{Annual\ Salary\ (Millions)}$$

While this provides a baseline production ratio, it suffers from a fundamental financial flaw: dividing total on-court output by salary yields a gross ratio that cannot produce negative values. In corporate finance, Net Return on Investment (ROI%) is defined as:

$$Net\ ROI\ =\ \frac{Value\ \mathrm{Pr}o\,duced\ -\ Cost}{Cost}\cdot 100$$

To measure contractual efficiency rather than raw output, the model must allow for negative returns when a player fails to justify their capital outlay. To operationalize this, we must establish what “Fair Market Value” looks like across the league.

Using representative starter averages, an everyday NBA starter plays roughly 70 games, logs 30 MPG, posts a 10.0% PIE (0.10 in decimal form), and earns approximately $20M annually:

$$Baseline\ Ratio\ =\ \frac{70\cdot 30\cdot 0.10}{20}\ =\ 10.5\ \approx\ 10.0$$

A score of 10.0 under the prior gross model represents the break-even baseline for an average contract.

Benchmarking against this 10.0 market equilibrium yields the final model: the **Contract Surplus Yield (CSY)** formula. By measuring percentage deviation from break-even production, CSY quantifies whether an investment yielded surplus value or net deadweight loss:

$$Contract\ Surplus\ Yield\ (CSY)\ =\ \left(\frac{Total\ Impact\ ROI\ -\ 10.0}{10.0}\right)\ \cdot\ 100$$

Bearing this in mind, we can create a scale to reference:

- **Tier 1: Toxic Assets / Dead Money** (CSY < −75%)

  Complete disasters, buyouts, and unamortized waivers (−100%)

- **Tier 2: Inefficient Contracts** (−75% to −40%)

  Substantial overpays for modest starter production

- **Tier 3: Below-Average Value** (−40% to −10%)

  Mild overpays coupled with minor durability or aging concerns

- **Tier 4: Fair Market Value** (−10% to +15%)

  Expected market production commensurate with compensation

- **Tier 5: Surplus Value** (CSY > +15%)

  High-bargain assets generating massive surplus production

Data Collection and Preparation

To evaluate these contracts empirically, I compiled an observational panel dataset tracking player-seasons across the 2020s. The dataset captures standard performance indicators alongside PIE and contract hits spanning the 2020–21 through 2025–26 seasons.

Translating raw sports data into an econometric model requires addressing critical edge cases:

1. **Text-to-Numeric Standardization:** Contractual figures are frequently published as delimited text strings (e.g., “$25,550,814”). Using Python’s

pandas library, these strings were stripped of dollar signs and commas, converted to floating-point numbers, and scaled to millions of dollars ($M).

2. **Accounting for "Dead Money" (Buyouts and Waivers):** A frequent flaw in basketball analytics is the automatic dropping of player-seasons with zero games played (GP = 0). In our dataset, Deandre Ayton (2025–26, $25.55M), Bradley Beal (2025–26, $96.92M), and Dāvis Bertāns (2024–25, $5.25M) recorded zero on-court minutes following contract buyouts or waiver releases. Omitting these rows would falsely imply that the franchise's financial liabilities vanished. These player-seasons were explicitly retained, imputing games, minutes, and PIE as 0. Under the CSY framework, this yields a complete total loss of −100.0%, capturing the true economic drag of dead cap space.
3. **Deliberate Omission of Future Contracts:** While Burns's original compilation featured sixteen contracts, Trae Young and Gary Trent Jr. were deliberately excluded from the empirical calculations. At the time of this writing (preseason, October 2026), both players have yet to log a single regular-season minute under their newly inked deals. Evaluating them without completed game data would be purely speculative; thus, the active study focuses on the fifty-two completed player-seasons across the remaining fourteen contracts.
4. **Multi-Column Prioritization:** A percentage metric produces an exact tie at −100.0% among all buyout players. However, from a managerial standpoint, losing $5.25M waiver is fundamentally different from losing 100% on a $96.92M buyout. To reflect this, our sorting logic applies a primary ascending sort on CSY

and a secondary descending sort on annual salary. Furthermore, I constructed an auxiliary metric, Wasted Salary ($M):

$$Wasted\ Salary\ (M)\ =\ Salary\ (M)\ \cdot\ \left(\frac{-CSY}{100}\right)$$

This metric directly calculates franchise deadweight loss in raw currency terms.

Empirical Findings and Analysis

Running the model across fifty-two player-seasons reveals significant stratification, confirming that not all bad contracts fail in the same manner.

Player	Season	Team	Salary ($M)	Wasted ($M)	GP	PIE	Net ROI (%)	Tier
Bradley Beal	2025-26	PHX	$96.92M	$96.92M	0	0.0	–100.0%	Tier 1: Toxic
Deandre Ayton	2025-26	POR	$25.55M	$25.55M	0	0.0	–100.0%	Tier 1: Toxic
Dāvis Bertāns	2024-25	CHA	$5.25M	$5.25M	0	0.0	–100.0%	Tier 1: Toxic
Evan Fournier	2023-24	NYK	$11.70M	$11.65M	3	1.4	–99.5%	Tier 1: Toxic
Dāvis Bertāns	2023-24	OKC	$10.55M	$10.46M	15	9.2	–99.1%	Tier 1: Toxic
Joe Harris	2023-24	DET	$19.93M	$19.56M	16	2.2	–98.1%	Tier 1: Toxic
Evan Fournier	2022-23	NYK	$18.00M	$15.89M	27	4.6	–88.3%	Tier 1: Toxic
Joe Harris	2021-22	BKN	$17.36M	$14.48M	14	6.8	–83.4%	Tier 1: Toxic
Dāvis Bertāns	2022-23	DAL	$16.00M	$12.62M	45	6.9	–78.8%	Tier 1: Toxic
Dāvis Bertāns	2021-22	WAS	$10.57M	$8.07M	34	5.0	–76.4%	Tier 1: Toxic
Zach Lavine	2023-24	CHI	$40.06M	$30.46M	25	11.0	–76.0%	Tier 1: Toxic

1. Tier 1 (Toxic Assets): The Mechanics of Cap Paralysis

The data strongly validates Lucas Burns's assertion that certain deals represent true "death sentences." Sitting alone at the absolute apex of contractual ruin is Bradley Beal's 2025–26 buyout with the Phoenix Suns. Absorbing a staggering $96.92M cap hit over multiple seasons with zero on-court contribution represents the single largest dead-money liability in modern league history.

Crucially, Tier 1 is not exclusive to buyouts; it also captures the catastrophic decline of active players sidelined by injury or demotion:

- **Evan Fournier with the New York Knicks (2023–24):** Logging just 3 appearances with a meager 1.4 PIE while earning $11.70M yielded a −99.5% CSY.
- **Joe Harris with the Detroit Pistons (2023–24):** After recurring ankle surgeries eroded his mobility, Detroit paid $19.93M for 16 games and a 2.2 PIE (−98.1% CSY).
- **Zach LaVine with the Chicago Bulls (2023–24):** Unlike specialist shooters, LaVine consumed maximum cap space ($40.06M). Appearing in only 25 games produced a devastating −76.0% CSY, representing over $30.46M in net wasted cap room.

2. Tier 2 (Inefficient Contracts): Durability Without Needle-Moving Production

While Tier 1 is driven by non-availability, Tier 2 represents the subtler trap in roster construction: paying superstar money to players who remain largely healthy but provide replacement-level or league-average output.

- **Paul George in Philadelphia (2024–25 & 2025–26):** Earning $49.21M and $51.67M respectively, George logged 41 and 37 games with above-average individual efficiency (~11.0 PIE). However, against a $50M baseline, his return sat at −71.3%

and −74.3%, demonstrating how ultra-max contracts severely penalize even moderate availability dips.

- **Bradley Beal in Phoenix (2024–25):** Playing 53 games and posting a 9.3 PIE against a $50.20M paycheck yielded a −68.5% CSY. Beal played, but each unit of production came at nearly triple the expected market cost.

3. Tiers 3 and 4: Validating the “Good Players Overpaid" Hypothesis

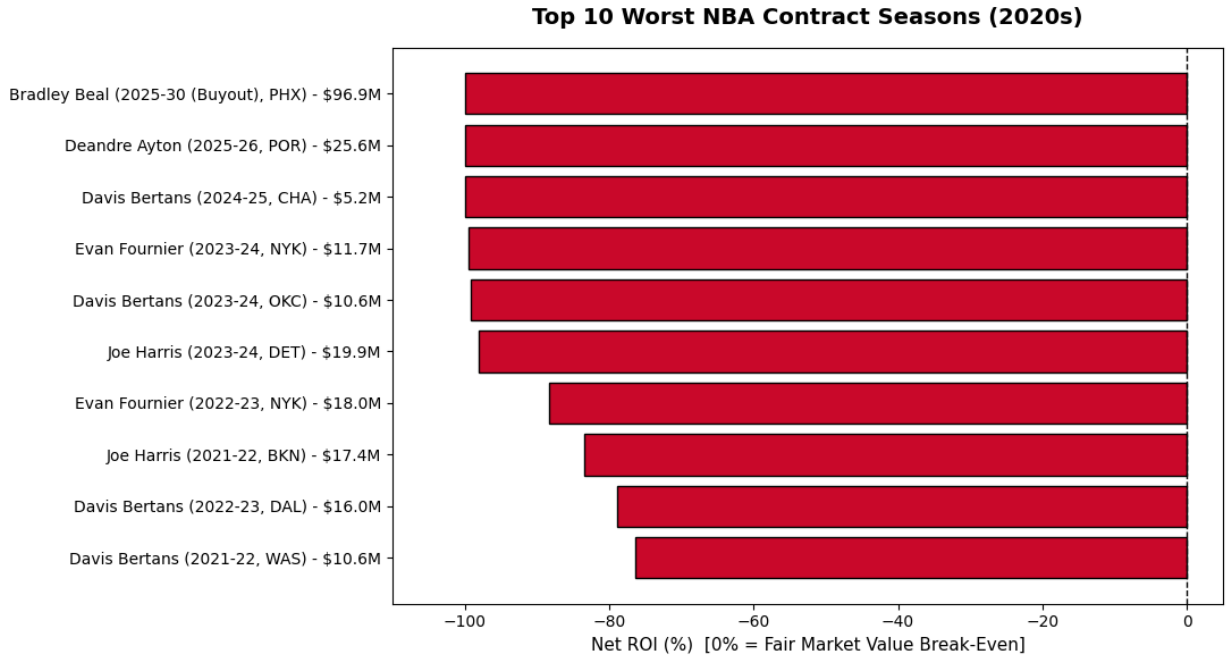
Perhaps the most compelling finding of the model is that it validates Burns's second claim: several contracts labeled as “disasters” were actually fair or above-average investments during their initial years before aging into inefficiency:

- **Kyle Kuzma (2023–24, WAS):** Earning $25.57M over 70 games with an 11.4 PIE produced a yield of +1.7%, placing him squarely in **Tier 4: Fair Market Value**.
- **Evan Fournier (2021–22, NYK):** In his first season with New York, Fournier played 80 games, set the franchise record for single-season three-pointers, and posted a +1.9% CSY against a $17.14M salary.
- **Joe Harris (2020–21, BKN):** Prior to injury, Harris led the NBA in three-point percentage over 69 games on a $16.07M deal, generating a +13.1% CSY—the highest single-season mark in the study by a wide margin.

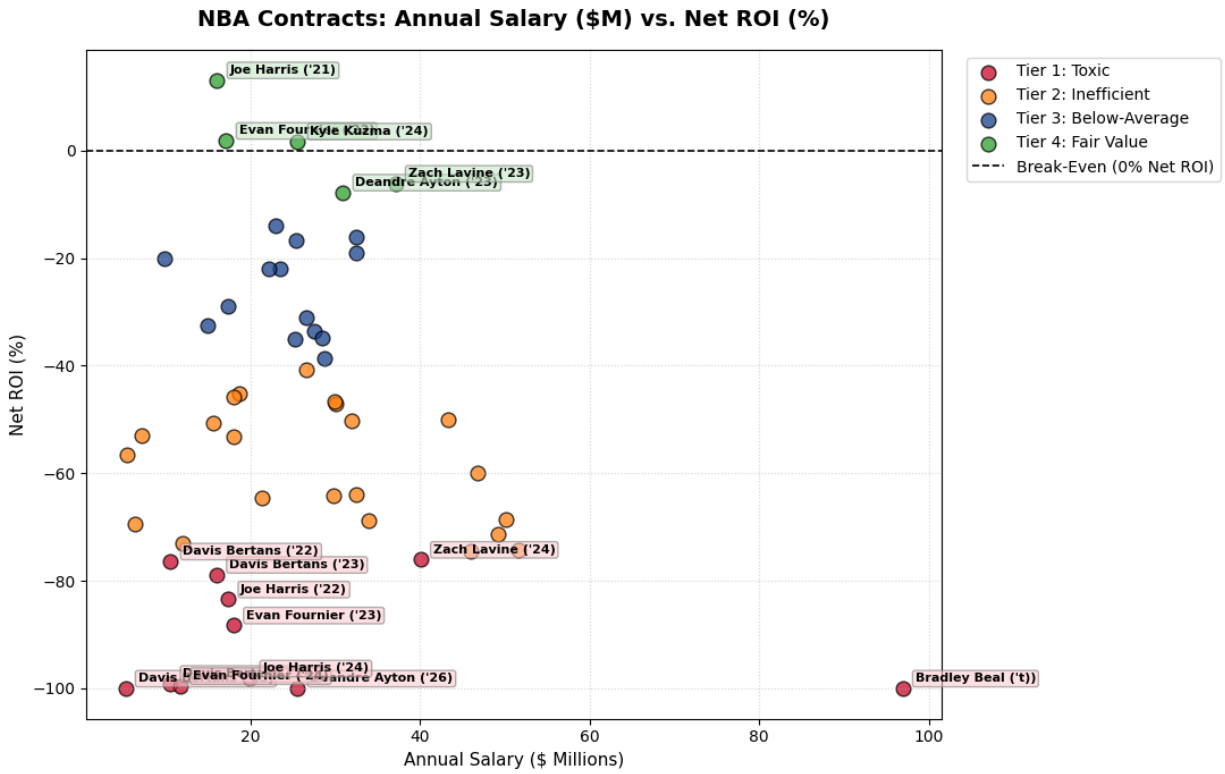
These contracts were not inherently toxic at signing; rather, they suffered from non-linear athletic decay and the compounding rigidity of guaranteed NBA contract structures.

Visualizing the Contract Matrix

To visualize these dynamics, I rendered two diagnostic charts using Pythons matplotlib framework:



1. **Figure 1 (The Wall of Shame):** The horizontal bar chart immediately isolates the severity gap among top-tier mistakes. While tiers are assigned via percentages, the magnitude of the financial liability makes it abundantly clear that Phoenix's buyout of Beal and Portland's buyout of Ayton operate in an economic stratosphere of their own.



2. **Figure 2 (Contract Matrix):** Plotting Annual Salary (X-axis) against CSY (Y-axis) exposes the asymmetric risk inherent in maximum contracts. The plot displays a distinct downward bifurcation:

- **“Availability Cliff” (Bottom-Left to Bottom-Center):** Players who suffer injuries or fall out of rotations drop straight down to the −100% baseline regardless of salary size.
- **The "Max Contract Trap" (Bottom-Right Quadrant):** When a $40M+ player suffers even a partial decline in availability or efficiency, they are pulled into the lower-right quadrant. Because the denominator is enormous, the financial deadweight loss extends far deeper into franchise cap space than lower-salary mistakes.

Under these constraints, a contract like Bradley Beal's −68.5% to −100% yield hard-caps a franchise's strategic flexibility and halts trade mechanics entirely. Measuring contractual efficiency has become a prerequisite for institutional survival in front office management and is no longer an academic novelty for sports fans.

Limitations

While the CSY model provides clear empirical grounding for subjective debates, several economic limitations must be acknowledged:

1. **Contextual Independence of PIE:** Although PIE comprehensively tracks individual box-score performance, it remains an unadjusted rate metric. It does not control for teammate quality, opposing lineups, or offensive scheme spacing as effectively as regression-based impact metrics like Estimated Plus-Minus (EPM) or DARKO.
2. **Static Break-Even Threshold:** Our model utilizes a fixed benchmark of 10.0 based on median starter production. In reality, market equilibrium shifts annually as national television revenues increase the salary cap by up to 10% per season.
3. **Defense and Intangibles:** Rim protection, point-of-attack defense, and locker-room leadership are notoriously difficult to capture through box-score-derived metrics, occasionally undervaluing defensive stoppers on high salaries.

Discussion and Case Study: The Bradley Beal Precedent

When I first set out to investigate Lucas Burns's Instagram rankings, I wondered whether empirical economic modeling could truly quantify the subjective pain of a bad

NBA contract down to the dollar. By shifting from a raw production ratio to a grounded CSY framework, the numbers revealed a compelling story.

Consider Bradley Beal's 5-year, $251M maximum contract originally signed with the Washington Wizards on July 6, 2022. Even now, the strategic rationale behind Phoenix's decision to trade for Beal remains perplexing. His last All-Star campaign was the 2020–21 season, when he averaged 31.3 PPG and earned All-NBA Third Team honors. The warning signs, however, emerged the following year when his scoring dropped from 31.3 to 23.2 PPG on virtually identical starter minutes (~35.9 MPG). Even so, Washington still awarded him a supermax extension with a full no-trade clause.

The Suns believed they were acquiring a dynamic third star to pair alongside franchise cornerstone Devin Booker and all-time great Kevin Durant. Yet they failed to consider roster equilibrium: the trio represented redundant, ball-dominant perimeter scorers sharing similar defensive limitations. Phoenix overlooked positional fit, defensive versatility, and, above all, long-term monetary flexibility—the ability to build a cohesive roster rather than over-indexing on three ball-dominant stars.

Name recognition alone does not guarantee winning production. To coaches, front-office executives, and fans alike, the pedigree of a "once-upon-a-time" star does not guarantee on-court surplus value. While I am fully in favor of athletes maximizing their earnings, lines must be drawn from a roster-building perspective.

A player's past brand does not guarantee future results. While NBA front offices are increasingly aware of this, the average fan remains drawn to nostalgia. Despite unprecedented access to advanced media and sports economics, public discourse often values familiar names over tangible statistical output.

Conclusion

In short, given the distinction between name and reputation, the former doesn't signify much of anything. As an outsider looking in, I'd like to imagine that the average NBA front office is well aware of this. But the average NBA fan, however, seems to have been stuck in the past. Despite the most convenient widespread access to NBA media more than ever before, let alone the most exigent relevance of economics in sport, the average NBA fan remains stuck in a past where phonemes and nostalgia, perhaps, forever mean more than results, when the reality is that all of these factors must be considered.

Let us as fans of sport bear in mind the importance of [monetary] literacy, if not for the success of the game, then at least for ourselves. After all, it almost seems like now more than ever is the economic state of the world in question.

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